

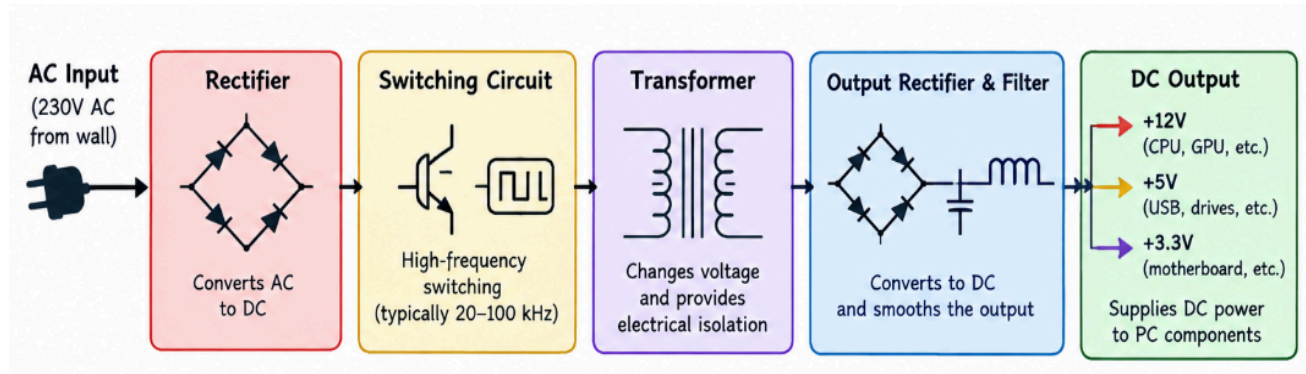
Assignment-2.2

“Powersupply system”

TASK-1 List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Supply Type	Device	WHY????
SMPS	Desktop Computer(PC)	Converts AC to multiple steady DC voltages (+3.3V, +5V, +12V) efficiently without excess heat.
SMPS	Laptop Charger brick	Provides high-efficiency power conversion in a compact, portable form factor.
SMPS	LED SmartTV	Delivers stable, regulated DC power to sensitive circuit boards and backlights.
UPS	Wi-fi router	Prevents internet drops during brief power cuts so online tasks aren't interrupted.
UPS	Desktop Computer	Provides emergency battery backup power to save work and shut down safely during outages.

TASK-2 Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. You can hand-draw and upload a photo, or use any free diagram tool like draw.



1.AC Input (110V / 220V): Draws raw alternating current from your wall outlet.

2.Primary Rectifier & Filter: Uses a diode bridge and large capacitors to convert high-voltage AC into high-voltage DC.

3.Switching Element (Transistors): Rapidly switches the DC voltage on and off thousands of times per second (kHz frequency) to regulate power efficiently.

4.High-Frequency Transformer: Isolates the high voltage and steps it down safely to low operating voltages.

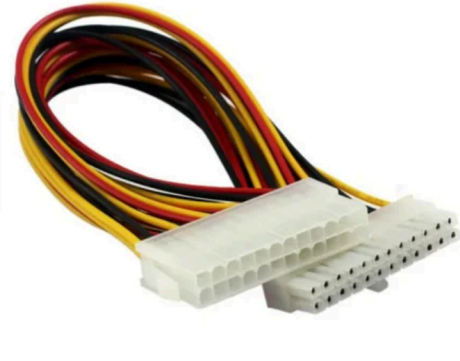
5.Output Rectifier & Filter: Smooths the high-frequency pulsed power back into clean, stable DC voltage.

6.DC Output Rails: Distributes regulated power directly to PC components via dedicated voltage lines:

- **+12V:** Powers high-draw components like the CPU and GPU.
- **+5V:** Powers storage drives (SSDs/HDDs) and USB ports.
- **+3.3V:** Powers motherboard logic chips and RAM.

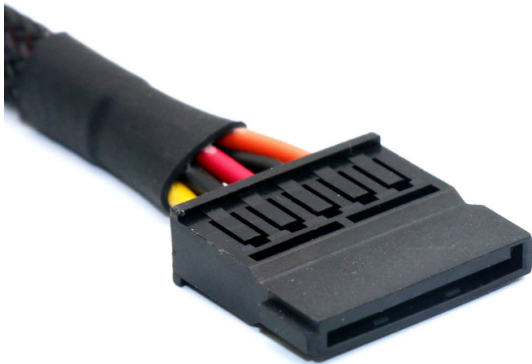
TASK-3 Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

1. ATX 24-Pin Motherboard Connector



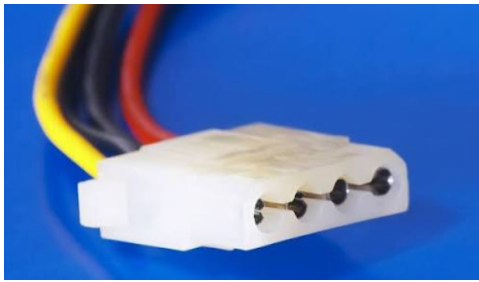
- **Devices Powered:** Main motherboard, RAM, PCIe slots, and onboard chipsets.
- **Safety Tip:** Hold the motherboard flat from underneath when pushing the connector in to avoid flexing and cracking the circuit board.

2. SATA Power Connector (15-Pin)



- **Devices Powered:** 2.5-inch/3.5-inch HDDs, SATA SSDs, and optical drives.
- **Safety Tip:** Align the L-shaped notch carefully; pulling or inserting at an angle can snap the plastic key on the drive.

3.Molex Connector (4-Pin)



- **Devices Powered:** Older hard drives, case fans, liquid cooling pumps, and RGB hubs.
- **Safety Tip:** Pins inside Molex plugs easily misalign; check pin alignment before applying pressure to avoid bending them.

4.PCie Power Connector (6-Pin / 8-Pin / 6+2-Pin)



- **Devices Powered:** Dedicated Graphics Cards (GPUs).
- **Safety Tip:** Ensure the safety latch clicks fully into place; loose high-draw GPU connections can melt cables.

TASK-4 Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

1. **Switch off and unplug the PC.**

Make sure the SMPS is disconnected from the wall socket before connecting any internal cables.

2. **Connect the motherboard.**

- Use the **24-pin ATX motherboard power connector**.
- Plug it into the large **24-pin ATX socket** on the motherboard.
- This supplies the main power to the motherboard.

3. **Connect CPU power.**

- Use the **4+4-pin EPS/CPU power cable** from the SMPS.
- Connect it to the **4/8-pin CPU power socket** near the CPU on the motherboard.
- This provides power to the CPU.

4. **Connect the graphics card (GPU).**

- If the graphics card requires additional power, use the SMPS's **PCIe power cable**, commonly **6+2-pin (8-pin) PCIe**.
- Some newer GPUs use a **16-pin 12V-2x6/12VHPWR connector**, depending on the GPU and PSU.
- Connect the appropriate cable firmly to the GPU.

5. **Connect storage devices.**

- For a **2.5-inch SATA SSD or HDD**, use a **SATA power connector** from the SMPS.
- Connect the SATA power cable to the drive.
- Remember that the drive also needs a **SATA data cable** connected between the drive and motherboard. The SATA data cable does **not** come from the SMPS.

6. **Check all connections.**

Make sure the connectors are fully seated and cables are routed safely away from fans.

7. **Connect the AC power cable last.**

Once the PC is assembled and the case is closed, connect the SMPS to the wall socket using its **AC power cable**.